FEYMAN

IM23AG

Hardware Integration Guide

Inertial Measurement Unit

Document Version: V1.4.3

Applicable Model: IM23AG

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Revision History

Version	Description	Date
V1.0	Initial release (sign-off: LQ)	01-2019
V1.1	Release (sign-off: LQ)	07-2020
V1.2	Release (sign-off: LQ)	01-2021
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V1.4.3	Release (sign-off: LQ)	05-2025

NOTE This guide targets hardware engineers integrating the IM23AG IMU module into a host platform. All port voltages are 3.3 V. Confirm pin assignments, reference-design values and mechanical dimensions against the released schematic and mechanical drawing before production.

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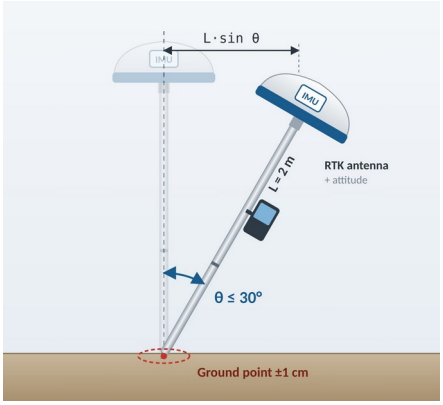
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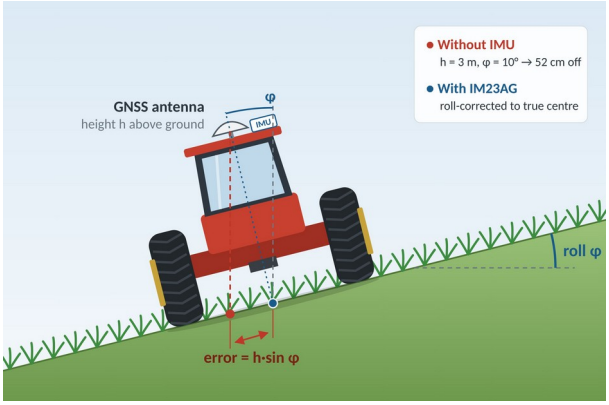
1 Introduction

The IM23AG IMU module fuses data from a MEMS accelerometer and gyroscope with GNSS RTK data to deliver high-precision attitude measurement, with roll and pitch accurate to within 0.05 degrees. This level of accuracy has found the IM23AG widespread use in industrial applications such as tilt RTK surveys, agriculture machine automation, and dead reckoning.

The IM23AG is protected under various international patents (US US11409001B2, Chinese CN110017849B, etc.).



Tilt RTK survey



Agricultural terrain compensation

2 Specifications

2.1 Accelerometer

ACCELEROMETER	
Range	±8 g
Bias accuracy	±5 mg

2.2 Gyroscope

GYROSCOPE	
Range	±1000 deg/s
Bias accuracy	±0.2 deg/s

2.3 Attitude

ACCURACY	
Attitude accuracy (roll / pitch)	±0.05 deg (95%)
Heading accuracy	±0.5 deg (95%)

2.4 Tilt Survey & Auto Steering

APPLICATION PERFORMANCE	
Tilt survey accuracy	1 cm, tilt ≤ 30°, 2 m pole
Tilt survey initialization time	1 second (95%)
Ag auto steering yaw accuracy	0.5° (95%)

3 Hardware Instructions

3.1 Electrical Characteristics

Parameter	Min	Typ	Max	Units
Supply voltage	3.0	3.3	3.6	Volt
Current consumption	120	130	140	mA

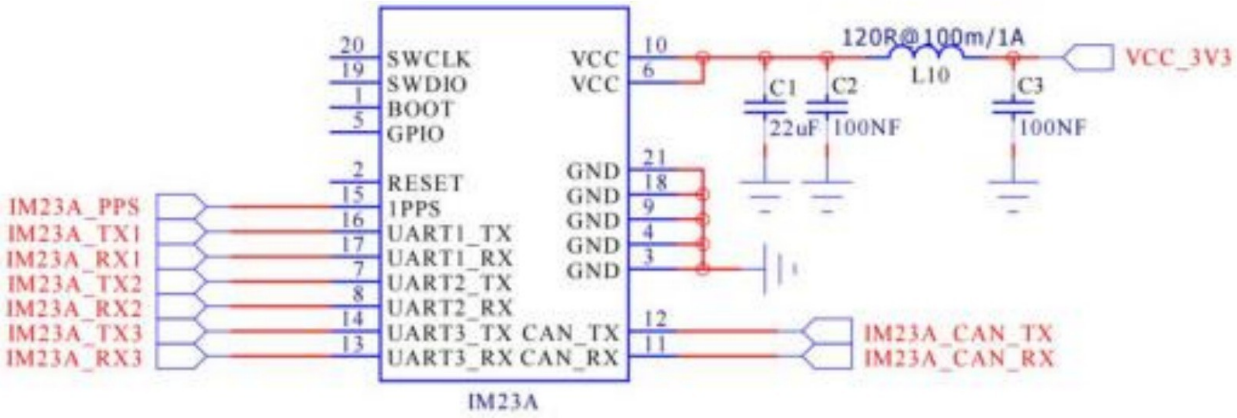
3.2 Pin Assignment

Pin	Name	Description (all port voltages are 3.3 V)
1	**BOOT**	Start modes — High: BOOT mode; NC or Low: normal working mode
2	**RESET**	Hardware reset pin (pull low 10 ms to reset module)
3	**GND**	Ground
4	**GND**	Ground
5	**GPIO**	Output IO; an active buzzer can be connected
6	**3V3**	Voltage supply
7	**UART2_TX**	UART2 output — connect to GNSS board RX
8	**UART2_RX**	UART2 input — connect to GNSS board TX
9	**GND**	Ground
10	**3V3**	Sensor power supply
11	**CAN RX**	CAN interface, data reception — connects to CAN transceiver
12	**CAN TX**	CAN interface, data transmission — connects to CAN transceiver
13	**UART3_RX**	UART3 input — configuration, message output and firmware upgrade
14	**UART3_TX**	UART3 output — configuration, message output and firmware upgrade
15	**1PPS**	1PPS pin — connect to GNSS board 1PPS
16	**UART1_TX**	UART1 output — configuration, message output and firmware upgrade
17	**UART1_RX**	UART1 input — configuration, message output and firmware upgrade
18	**GND**	Ground
19	**SWDIO**	Debug IO (reserved)
20	**SWCLK**	Debug IO (reserved)
21	**GND**	Ground

3.3 Reference Design

UART ports 1 and 3 have the same functions; one can be connected to the main processor and the other used as a debugging interface. In addition, the BOOT pin is recommended to be routed for backup. The buzzer can be used for status indication, such as power-on and completion of installation-angle estimation.

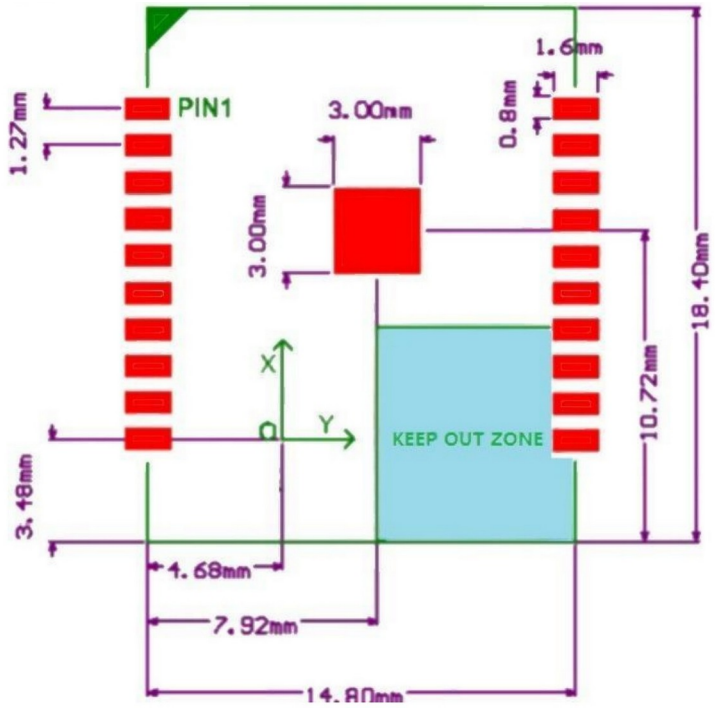
In the reference design, both VCC pins (6 and 10) are decoupled with 22 µF (C1) and 100 nF (C2) at the module and fed from VCC_3V3 through a 120 Ω @ 100 MHz, 1 A ferrite bead (L10), with a further 100 nF (C3) on the supply side. CAN TX and CAN RX (pins 12 and 11) connect to the host CAN transceiver.



IM23AG reference design

3.4 Mechanical

The module height is 3.2 ± 0.1 mm; the edge pad size is 1.6 mm × 0.8 mm and the middle pad size is 3 mm × 3 mm. It is recommended to place the IM23AG module in a thermally stable region to avoid convection with outside air, and external stress on the module is strictly forbidden. The coordinate system is defined by the right-hand rule, as shown below.



IM23AG land pattern and coordinate system (mm)

4 SMT Precautions

The IM23AG modules must be baked at low temperature to remove water vapor before SMT. Bake for at least 24 hours at 60 °C prior to SMT. For the detailed SMT temperature profile, refer to the reference document: FMI FEYMAN IMU SMT Requirements.pdf.